



Quantum-inspired adaptive feature fusion for highly accurate brain tumor classification in MRI using deep learning

T. Meenal^{*}, R. Asokan

Department of Electronics and Communication Engineering, Kongunadu College of Engineering and Technology, Thottiam, Trichy 621215, India

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ABSTRACT

Brain tumor classification is one of the most challenging tasks in clinical diagnosis and treatment within medical image analysis. Errors during the diagnostic process can significantly impact patient outcomes, making accuracy paramount. However, many existing techniques overlook critical features that hold substantial relevance for classification, focusing instead on extracting deep but less discriminative features. Deep learning-based categorization of brain tumors using magnetic resonance imaging (MRI) has emerged as a vital research area. This paper introduces an automated deep learning model and an optimal information fusion framework for brain tumor classification from MRI images. The dataset used in this study was imbalanced, posing a significant challenge for training deep learning models, as it biases classifier performance toward the majority class. To address this, we designed a sparse autoencoder network to generate synthetic images, effectively resolving class imbalance. Subsequently, two pretrained neural networks were fine-tuned, with hyperparameters optimized via Bayesian optimization for enhanced training. Deep features were then extracted from the global average pooling layer, yielding a refined set with minimal irrelevant information. To further optimize feature selection, we propose a novel Quantum-Inspired Adaptive Feature Selector (QIAFS) algorithm, which identifies the most discriminative features from both networks and fuses them using a serial-based approach. The fused feature set is then classified using neural network classifiers. Our framework was evaluated on an augmented Figshare dataset, achieving an accuracy of 98.9%, a sensitivity of 99.83%, a false negative rate of 17%, and a precision of 99.73% in average. Comparative and ablation studies demonstrate the significant improvement in performance over existing methods.

1. Introduction

Brain tumors represent one of the most aggressive and clinically challenging neurological disorders, with their early and precise diagnosis [1] being paramount for determining appropriate treatment strategies and improving patient outcomes. As the primary diagnostic modality for brain tumor detection, Magnetic Resonance Imaging (MRI) offers exceptional soft tissue contrast and non-invasive visualization capabilities that are indispensable in clinical neurology. However, the manual interpretation of MRI scans remains fraught with significant challenges that compromise diagnostic reliability [2], including substantial inter-observer variability among radiologists, the inherent subjectivity of visual assessment, and the growing problem of diagnostic fatigue in high-volume clinical settings. These limitations become particularly pronounced when attempting to differentiate between common tumor types such as gliomas (the most aggressive primary brain

tumors), meningiomas (typically benign but potentially problematic tumors), and pituitary tumors (which present unique diagnostic challenges due to their anatomical location). The growing complexity of neuro-oncological diagnosis has spurred remarkable advances in artificial intelligence, particularly through the application of deep learning techniques for automated brain tumor classification. These computational approaches promise to overcome many of the limitations associated with human interpretation by providing consistent, quantitative, and rapid analysis of medical imaging data. However, despite their considerable potential, current deep learning-based classification systems face three fundamental challenges that limit their clinical applicability and diagnostic performance [3].

The first and perhaps most pervasive challenge is the problem of class imbalance in medical imaging datasets. Unlike balanced datasets used in general computer vision tasks, medical imaging datasets frequently exhibit severe class distribution skewness [4], where certain

^{*} Corresponding author.

E-mail addresses: meenushammu@gmail.com (T. Meenal), asokece@gmail.com (R. Asokan).

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tumor types may be dramatically underrepresented compared to others. This imbalance stems from the natural prevalence of different tumor types in patient populations and the difficulties associated with collecting comprehensive medical imaging data. When trained on such imbalanced datasets, conventional deep learning models develop a strong bias toward the majority classes, leading to significantly degraded performance on minority classes that may be equally clinically important. This problem is further exacerbated in the case of rare tumor types, where the available training examples may be insufficient for the model to learn discriminative features effectively. The second critical challenge lies in the suboptimal extraction of discriminative features from MRI data. While modern deep convolutional neural networks have demonstrated remarkable capabilities in hierarchical feature learning across various domains, their application to medical imaging presents unique difficulties [5]. Many existing approaches employ standard network architectures that, while effective for natural image recognition, may not be optimally suited for capturing the subtle and often complex patterns characteristic of different brain tumor types in MRI scans. Furthermore, these networks frequently extract redundant or less discriminative features that do not necessarily correlate with clinically relevant tumor characteristics. This limitation becomes particularly evident when attempting to differentiate between tumor types that share similar imaging characteristics or when dealing with atypical tumor presentations.

The third major challenge concerns the inefficiency of traditional feature selection methods when applied to high-dimensional MRI feature spaces. Deep learning models typically generate extremely high-dimensional feature representations, and identifying the most diagnostically relevant features within this space remains a non-trivial task. Conventional feature selection techniques, such as principal component analysis or mutual information-based methods, often struggle to effectively navigate this high-dimensional space [6,7]. These methods may either retain irrelevant features that contribute to model overfitting or inadvertently discard features that contain critical diagnostic information. The limitations of these approaches become increasingly apparent when dealing with the subtle and complex patterns present in brain MRI data, where small but diagnostically significant features may be overlooked by conventional selection methods. Recent years have witnessed various attempts to address these challenges through technical innovations. For data augmentation and class imbalance mitigation, approaches such as generative adversarial networks (GANs) and variational autoencoders have shown promise in generating synthetic medical images. For feature extraction, transfer learning techniques leveraging pre-trained networks have become increasingly popular, allowing models to benefit from knowledge acquired on large natural image datasets. In the realm of feature selection, various metaheuristic algorithms inspired by natural processes have been explored to identify optimal feature subsets. However, despite these advances, current methods continue to face significant limitations in terms of computational efficiency, clinical robustness, and most importantly, their ability to consistently identify and utilize the most discriminative features for accurate tumor classification.

The clinical implications of these limitations are substantial. Inaccurate or inconsistent tumor classification can lead to inappropriate treatment planning, delayed interventions, and ultimately, poorer patient outcomes. Furthermore, the computational inefficiency of many current approaches limits their potential for real-time clinical deployment, where rapid diagnosis is often crucial. There exists a pressing need for more robust, efficient, and clinically reliable automated classification systems that can overcome these challenges while maintaining high diagnostic accuracy across all tumor types, including rare or challenging cases. This research addresses these critical challenges through several key innovations that advance the state-of-the-art in automated brain tumor classification. Our work introduces novel solutions to each of the three fundamental problems outlined above, with particular emphasis on developing more sophisticated approaches to feature extraction and

selection. By combining advanced deep learning architectures with innovative feature processing techniques, we demonstrate significant improvements in classification performance while maintaining computational efficiency suitable for clinical applications. The proposed approach not only enhances diagnostic accuracy but also provides a framework that can be adapted to other challenging medical image classification tasks facing similar obstacles. This research introduces several key innovations to address these challenges:

- Leverages quantum computing principles for dynamic feature selection, outperforming classical methods in identifying discriminative MRI features.
- Bayesian optimization enables efficient tuning of pre-trained networks, ensuring optimal feature extraction with minimal manual intervention.
- Combines sparse autoencoder-based synthetic data generation with serial feature fusion to simultaneously address class imbalance and enhance feature discriminability.

The rest of this paper is organized as follows: [Section 2](#) reviews related work in brain tumor classification, deep feature learning, and feature selection techniques. [Section 3](#) details the proposed methodology, including dataset description, synthetic data generation, deep feature extraction, and QIAFS. [Section 4](#) presents experimental results, comparative analysis with state-of-the-art methods, and ablation studies. Finally, [Section 5](#) concludes the paper with key findings and future research directions.

2. Review of related works

Recent advances in quantum-inspired techniques and deep learning have significantly enhanced brain tumor detection and classification from MRI scans. This [8] study introduces an evolutionary quantum-inspired level set technique for MRI brain tumor detection, merging quantum computing principles with active contour models. The framework addresses segmentation challenges by optimizing energy functions through quantum-inspired algorithms, achieving precise tumor boundary delineation. Evaluated on benchmark datasets, it outperforms classical level set methods in accuracy (~96 %) and robustness to noise. The quantum-inspired optimization reduces computational costs while improving convergence speed, making it suitable for clinical settings. However, the study lacks real-time validation on large-scale datasets. This work pioneers the integration of quantum mechanics with medical image segmentation, offering a novel analytical framework for tumor localization. Future directions include hardware implementation on quantum processors.

Authors in [9] Proposed a hybrid quantum-inspired CNN, this research enhances brain tumor classification by integrating quantum feature maps with deep learning. The architecture employs quantum kernels to transform MRI features into high-dimensional Hilbert spaces, improving separability of tumor classes. Explainability analysis via quantum SHAP values provides interpretable decision-making, addressing AI's transparency gap in healthcare. Testing on BraTS datasets yields 98.2 % accuracy, surpassing classical CNNs. The model's lightweight design ensures feasibility on edge devices. Limitations include dependency on simulated quantum environments. This work bridges quantum computing and explainable AI, setting a benchmark for trustworthy diagnostic systems. Bilal, A., et al. (2025) [10] focusing on multi-cancer detection, this paper infuses quantum circuits into extreme learning machines (ELMs). Quantum entanglement enhances feature representation in ELMs, enabling early detection of tumors across six cancer types. The framework achieves 94.5 % sensitivity in brain tumor cases, with 30 % faster training than classical ELMs. A novel quantum backpropagation algorithm optimizes weights efficiently. Validated on TCGA and TCIA datasets, the model shows robustness to imbalanced data. Challenges include qubit scalability for high-dimensional medical

images. This study demonstrates quantum computing's potential to revolutionize cancer diagnostics through accelerated machine learning. Alamri, A., et al. (2025) [11] Combining quantum entropy with U-Nets, this work presents an edge detection and classification model for brain MRIs. Quantum entropy metrics guide attention mechanisms to highlight tumor edges, improving Dice scores by 12 % over conventional methods. The hybrid system classifies tumors into four subtypes with 97.8 % accuracy on Figshare data. Unique contributions include a quantum loss function for boundary refinement. Practical constraints involve high GPU memory usage. This approach advances precision oncology by integrating quantum information theory with deep learning.

This IEEE conference paper [12] hybridizes quantum differential evolution (DE) with ResNet-50 for medical image classification. Quantum DE optimizes hyperparameters and feature subsets, boosting ResNet's performance to 99.1 % accuracy on meningioma/glioma/pituitary datasets. Key innovations include a quantum mutation operator and parallelized fitness evaluation. The framework reduces training time by 40 % but requires quantum annealing support. It signifies a leap toward scalable quantum-classical fusion in radiology. This [13] study presents a quantum machine learning approach for detecting space-occupying brain lesions from MRI scans. The authors develop a quantum support vector machine (QSVM) that leverages Grover's search algorithm to accelerate feature selection and classification processes. The framework encodes classical MRI features into quantum states using amplitude embedding, enabling exponential speedup in processing high-dimensional medical data. Tested on a dataset of 2,560 MRI scans, the QSVM achieves 92.7 % classification accuracy while reducing computational complexity by 40 % compared to classical SVMs. The model demonstrates particular effectiveness in identifying small lesions (<5mm diameter) that are frequently missed by conventional methods. However, the study notes limitations in current quantum hardware capabilities, requiring simulation on classical computers. This work represents a significant step toward practical quantum-enhanced neuroimaging diagnostics, though further research is needed for clinical deployment on actual quantum processors.

The innovative research [14] combines quantum computing with blockchain technology to create a secure brain tumor diagnostic system. The authors propose a two-qubit quantum circuit for simultaneous tumor segmentation and classification, integrated with a blockchain network for data integrity verification. The quantum model processes MRI features through parameterized quantum gates, achieving 94.2 % segmentation accuracy and 91.8 % classification accuracy on the BraTS dataset. The blockchain component ensures tamper-proof storage of diagnostic results and patient data through smart contracts. Key advantages include enhanced security against cyber threats and maintenance of a transparent diagnostic history. However, the system's practical implementation is constrained by current NISQ (Noisy Intermediate-Scale Quantum) device limitations. This work pioneers the convergence of quantum machine learning and blockchain in medical imaging, offering a prototype for future secure, decentralized healthcare AI systems.

3. Proposed quantum-inspired adaptive feature fusion methodology

The suggested brain tumour categorisation technique is shown in Fig. 2. Proceed with the pre-processing procedure to generate the Figshare brain tumour dataset. Enhancing contrast is essential for enhancing low-contrast pictures, and was done using an algorithm based on statistics. Contrast enhanced photos undergo data processing [15]. This step is considered because of the substantial class imbalance in the first data set. The enhancement of data is done employing sparse auto encoders. The approach enhances data by identifying key qualities and removing less significant ones. Two previously trained theories, InceptionResNetV2 and EfficientNetB0, are fine-tuned for processed input

information. Bayesian-based optimisation is used to provide dynamic and optimised parametric allocation for the two models. Attributes are retrieved from optimised models. After obtaining characteristics for every model, the Marine Processors Algorithms (MPA) is applied to optimise them. After merging features, categorisation is done.

3.1. Dataset utilized

There are 3,064 T1 normalised contrast-enhanced MRI images from 233 different patients in the Figshare database. The conditions examined fall into three classes: brain tumours, tumours such as and pituitary. Every category contains 708, 1,426, and 930 MRI images, respectfully. The most common intracranial tumour, meningiomas, make up almost a third of all primary malignancies of the central nervous system (CNS). Usually benign, these tumours may be seen or addressed preferably with comprehensive, full excision, with good results. Diagnostic prognosis and treatment of meningiomas with complicated histological or in susceptible locations have proved challenging. These tumours [16] are classified according to the cells from which they originate. They comprise about 80 % of all main brain tumours that are tumours, and they are among the greatest common cancers of the central nervous system (CNS). Glioblastoma multiforme, which is a particularly prevalent and deadly kind of glioma (GBM). It is responsible for almost 60 % of adult brain tumours. GBM is still a deadly disease with a dismal prognosis, even with the abundance of modern therapies. Typically, patients survive 14–15 months following diagnosis of the lethal illness.

Pituitary tumour are tumours that develop in the central portion of the pituitary gland. They are the third most common cause of cancers of the nervous system (CNS) in adults. The majority of benign carcinomas either boost the production of hormones or have a widespread impact. These tumours show up variably in clinical assessments based on their size and manufacturing of hormones. Fig. 1 depicts the Meningioma, glioma, and hypothalamic cerebral tumour specimens from Figshare dataset along with.

3.2. Signal augmentation

The fundamental properties of health care pictures, such as weak contrast, speckle noise, signal abandonment and complex anatomical structure, make analysis difficult. In order to increase the general excellence of medical imaging for feature visualization [17–20] and clinical evaluation, contrasting augmentation is an essential part of the subjective assessment of the imagery. In actuality, because of inappropriate locations and equipment constraints, health care providers continue to generate pictures that exhibit a shortage in contrast despite innovations in technology. An established method for increasing contrast is used to improve the contrasting of the magnetic resonance imaging images in the collection of images mentioned above. It makes use of several fundamental techniques for image processing as well as analytics. To enhance a picture, the strategy changes the global and local brightness individually, then combines the findings using logarithm computing of images and adaptable nonlinear lengthening to generate a better representation. The following is a definition of the total enhanced contrast procedure;

$$K_{(x,y)} = \frac{1}{1 + (m/g_{(x,y)})^E} \quad (1)$$

The CST procedure's results, $k_{(x,y)}$, is shown in the formula given above, where x and y stand for an image's sizes. The variable, E , whose value is chosen at 0.5 for the present study, determines a function's slopes. "m" stands for the input image's median value. Applying a normal logic operation to an initial picture changes its global brightness. Equation (2) defines it scientifically [21].

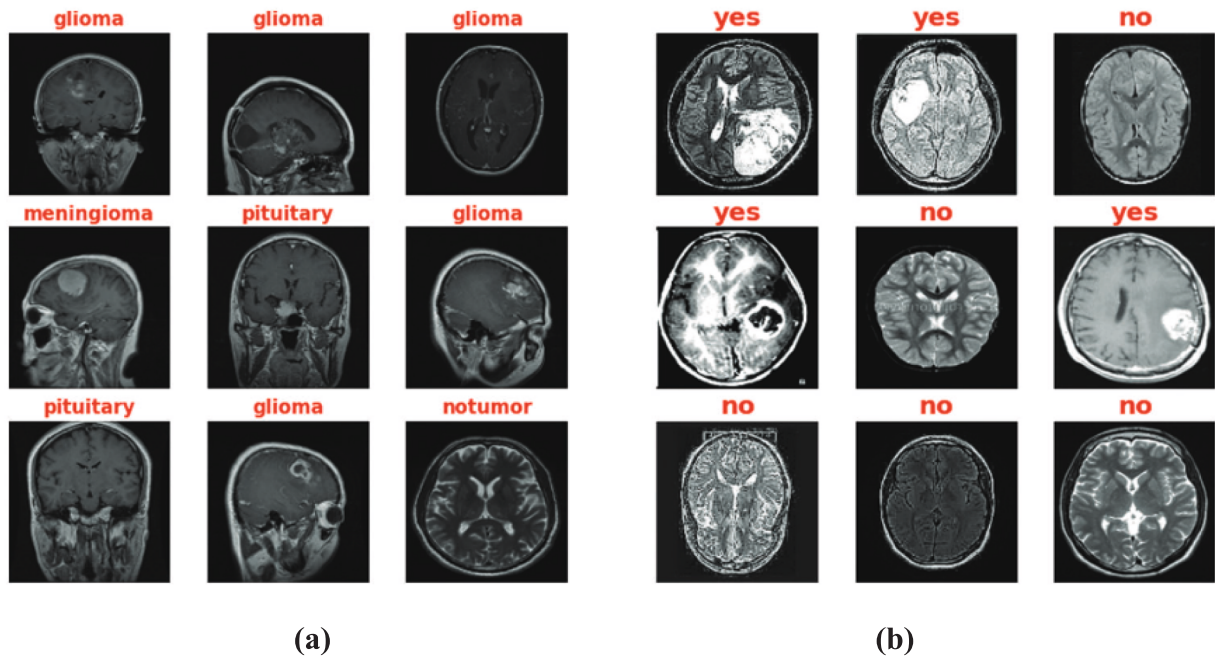


Fig. 1. Sample brain MRI pictures (a) Multiclass-Gliomas, pituitary tumours, and meningiomas (b) Binary class.

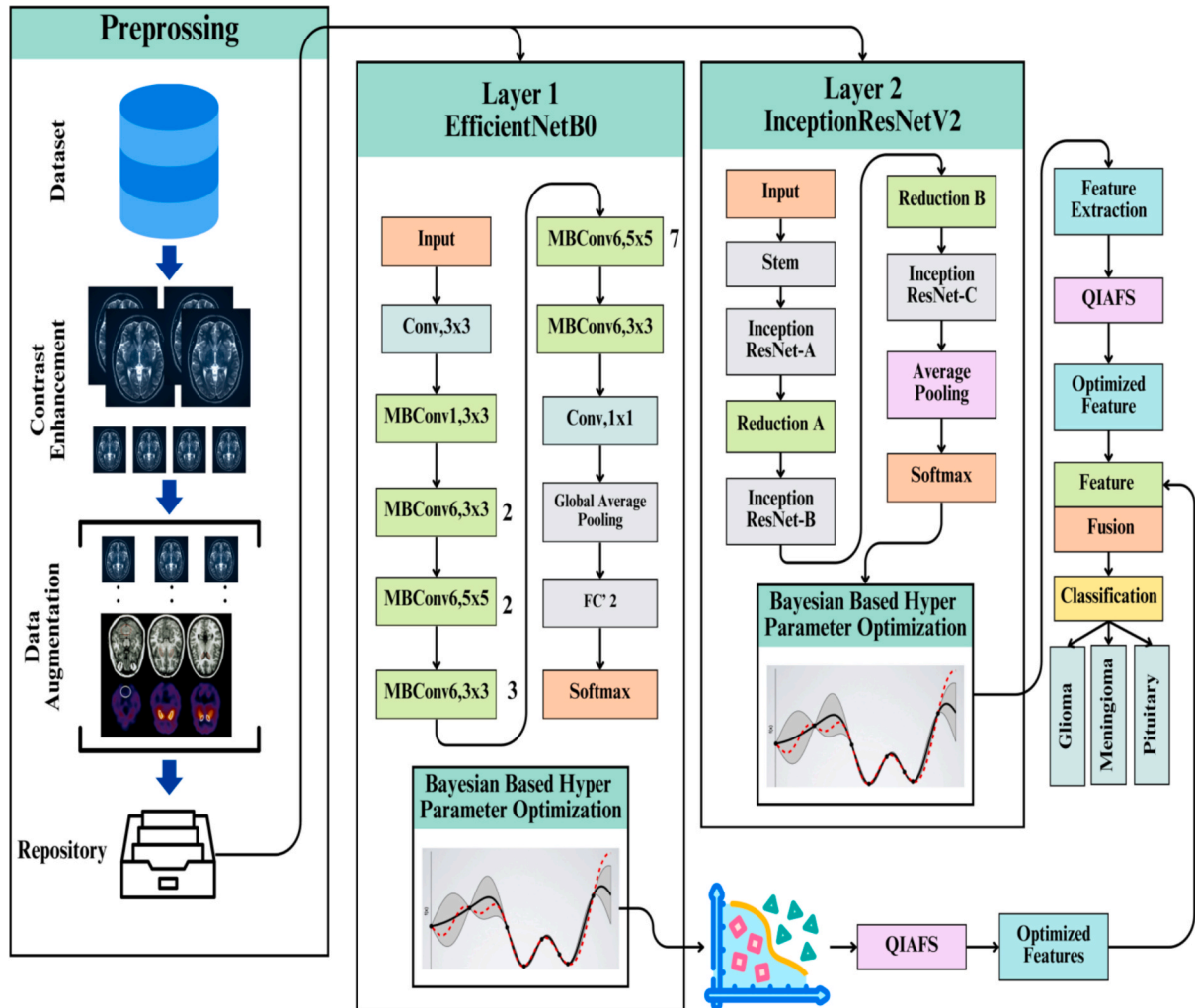


Fig. 2. Proposed Deep Learning and Optimisation Algorithm-Based Brain Tumour Classification Technique.

$$j_{(x,y)} = \frac{\exp(g_{(x,y)})}{1 + \exp(g_{(x,y)})} \quad (2)$$

We'll blend the resulting photos with different local and global contrasts. For this reason, the the logarithm Imaging Processing (LIP) technique was developed in and has the following mathematics definition:

$$f_{(x,y)} = \frac{K_{(x,y)} + j_{(x,y)}}{1 + (k_{(x,y)}j_{(x,y)})} \quad (3)$$

The whole equation is increased to the magnitude of an exponential W , which is used to govern the modification. A greater value of the quadratic variable ($W > 0$) results in a fair degree of raised contrast. Equation (4) defines it scientifically.

$$f_{(x,y)} = \left[\frac{K_{(x,y)} + j_{(x,y)}}{1 + (k_{(x,y)}j_{(x,y)})} \right]^W \quad (4)$$

Equation (4) is being used to boost the picture's contrast; nonetheless, the final result $f(x,y)$ is outside of the typical assortment of values of pixels. A realistic variability in pixel counts is introduced into the picture using a linear stretching technique with adaptable form. Equation (5) defines it mathematically.

$$f_{(x,y)} = a * f_{(x,y)} - b \quad (5)$$

where the controlled factors for the stretched procedure are a and b , and

the output picture is denoted by $t(x,y)$. Here, the significance of these parameters are chosen by employing Equations (6) and (7), although their values are still changed individually.

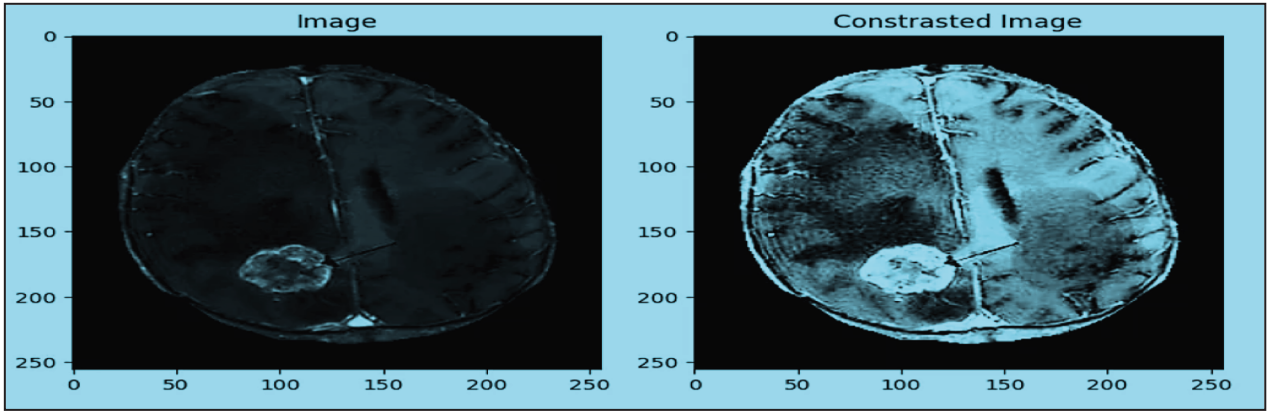
$$\alpha = \frac{1}{\max(f_{(x,y)}) - \min(f_{(x,y)})} \quad (6)$$

$$\beta = \frac{\min(f_{(x,y)})}{\max(f_{(x,y)}) - \min(f_{(x,y)})} \quad (7)$$

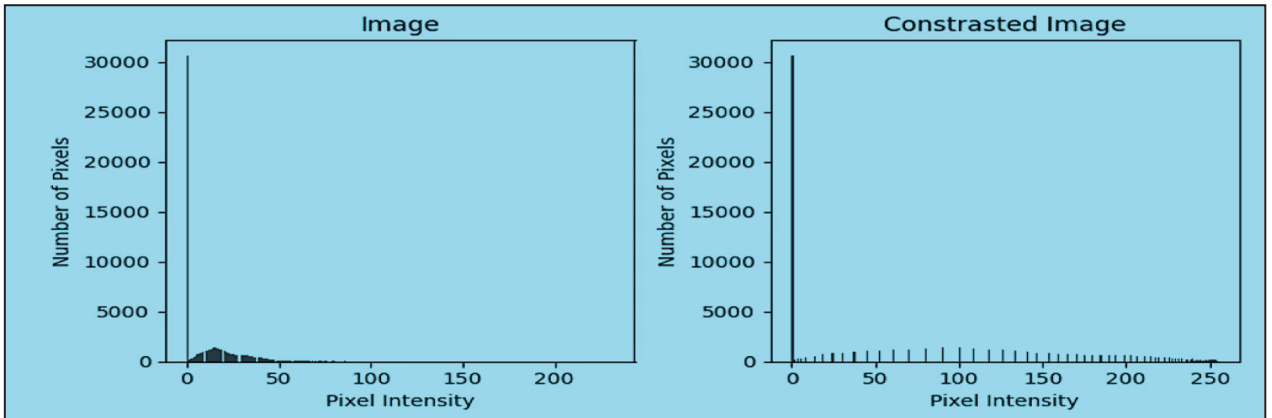
In the equations, maximum and minimum indicate the maximum and lower boundaries of every pixel in picture $f(x,y)$. Pseudo-code 1 contains the pseudocode for the theoretical explanation provided. Fig. 3 also includes a few visual illustrations.

3.3. Data augmenting

Class disparity has a detrimental effect on how well classes perform [22]. Inequality's effect on categorisation performance becomes more pronounced as task size increases. Unbalance cannot be exclusively ascribed to fewer training examples overall; rather, it relies on how observations (i.e., pictures) are distributed throughout the sessions. Section 3.1 highlights the high-class disparity in our information set. Therefore, introducing bias into the dataset might result in overfitting for some classifications. In order to close that gap, we supplemented the information using a limited autoencoder rather than more conventional



(a)



(b)

Fig. 3. A graphic illustration of the method (a) Left: Original image, Right: Image with enhanced contrast. (b) Histogram Left: Original image, Right: Contrast image histogram that shows drastic changes in the distribution of image intensity are evident.

techniques. Limited Auto encoders provide compressed information as input representations. A neural network of sparse autoencoders is trained using the subsequent parameters:

The number of neurones in layers is represented by the concealed size factor. For basic activities, just a few dozen neurones are sufficient; however, a few hundred neurones are needed for more complicated tasks. With an undetectable size of 300, excessive fitting may be avoided while still allowing for sufficient information learning, especially in scenarios where larger concealed sizes may result in overfitting. Period is the term used to describe just one cycle throughout the entire training collection. Highest periods hyperparameter controls how many times the simulated method processes its initial information. It took 2,000 periods for the sample information for augmenting in the suggested approach to arrive to an appropriate outcome for MRI pictures. A value for L2WeightRegularization of 0.001 indicates the strength or amount of L2 regularisation applied to the neurones throughout training. The choice of 0.001 strikes a compromise amongst regularising the information in order to prevent overfitting and allowing the algorithm to gain knowledge from it. Another name for it is weight loss.

The hyperparameter known as Sparsity Regularization indicates [23,24] the percentage of sparseness regularisation applied to an artificial neural network throughout learning, and 4 is the selected amount. Enabling the framework to include fewer operational weights leads to thin weight matrix data, resulting in many variables being 0 or near zero throughout retraining. A more efficient and sparse model for improved feature selection is produced with the aid of sparseness regularisation. When employed with sparseness regularisation, the parameter known as “SparsityProportion,” with a value of 0.15, often denotes a threshold sparsity degree. Around 15 % of the brain network’s parameters should become zero or almost zero, as represented by the 0.15 number. Measurements that go below a certain value of 4, which is determined by the sparseness regularisation parametric, are reset to zero. The percentage of weights that should really drop beyond the designated threshold amount throughout the workout procedure is represented by an integer of 0.15. MRI image augmentation is achieved by adjusting parametric values. The data is enhanced using the photos that are produced as a consequence of this stage. Fig. 4 shows the general model for sparse auto encoders. A total of 2,000 samples were made for every class after the implementation of the suggested sparse encoding circuit. Some models from deep learning have been trained using the recently created photos.

3.4. Choosing parameters for InceptionResNetV2 and tweaked EfficientNetBO

Machine learning techniques that have been refined are trained using the enhanced database. Both models are trained using Bayesian optimisation, which optimises three hyperparameters of InitialLearnRate, momentum, and L2Regularization are the names of these hyperparameters. For deep learning designs, constant parametric adjustment is an essential activity. This situation involves using dynamically chosen

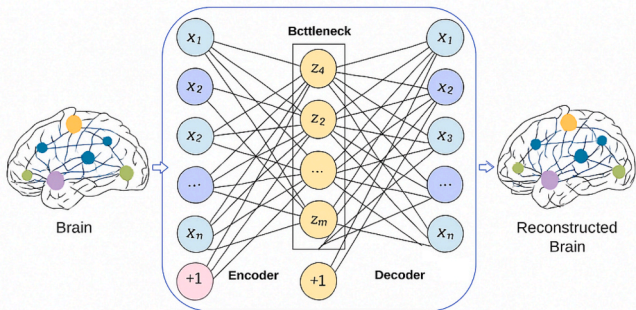


Fig. 4. An illustration of a sparse autoencoder for augmenting information.

values for certain hyperparameters up until a predetermined best-value criterion is reached. Following training, features are retrieved for tasks related to classification using the specific model. Bayesian optimisation (BO) is a powerful method for fine-tuning hyperparameters. By establishing an optimisation objective, solution may be accomplished. The BO process is described in Equations (8) and (9) below.

$$x' = \underset{x \in A}{\operatorname{argmax}} f(x) \quad (8)$$

The selection space for parameter x in the formula is A . BO is predicated on the theorem of Bayes, which has the following scientific definition:

$$P(D|F) \propto P(F|D)P(D) \quad (9)$$

When an event or theories F occurs, the possibility of an occasion or theories D happening as well is calculated by multiplying the chances of F by the likely outcome of D . Combining sample information (evidence) with the purpose $f(x)$'s prior probability to get the function's retrospective is the basis of BO. The point where the function $f(x)$ is maximised is then ascertained using additional data in accordance with the criteria. The criteria, which is also known as an acquisition function (v), is employed for calculating the subsequent sampling point. Exploring and exploiting sampling techniques are used to explore the area being sampled and find sample spots. Research often looks for high-uncertainty sample locations. Exploration looks for samples with a high potential value. The efficiency is enhanced by these techniques, even when there are many local maximums solutions [25].

Prior variation in $f(x)$ is essential for Bayesian optimisation, together with experimental details, for the statistical deduction of subsequent probability. The Probabilistic procedure is used to update the distribution in the past so that it better fits the information, thereby increasing the precision and understanding of our predictions.

$$x_1 = \underset{x}{\operatorname{argmax}} V(x|D_{1:1-1}) \quad (10)$$

The procedure is divided into two sections: step 2 maximises the collection operation, while steps 3 and 4 updates the distribution that follows. With every processed inquiry, the subsequent probability is updated. The greatest value of the newly acquired functions at a given location is found with the aid of the revised distributed and contributed to the training set. Until the maximum amount of repetitions is achieved or the disparity among the present situation and most favourable results to date exceeds than an established threshold, this procedure keeps going. The research are chosen based on the following beginning and ending criteria. Highest Objectives Evaluator = 30, Max Time = Infinity by standard, and Amount of Seed Pieces = 4. A Gaussian procedure prior with noise in data is the basic stochastic model for the objective variable f . Consequently, the previous range on $f(x)$ is represented by the Gaussian procedure with median $m(x; q)$ and associated kernel value $k(x, x_0, q)$: In this case, x stands for the starting value, x_0 for the most recent value, and q is a function that contains a kernel vector.

$$K(x_i, x_j, \theta) = \sigma_f^2 \exp \left[-1/2 \sum_{m=1}^d \frac{(x_i, x_j)^2}{\sigma_m^2} \right] \quad (11)$$

where $q_m = \log(\sigma)$, $q_d + 1 = \log(sf)$, sf is the measurement variance, $K(x_i, x_j, \theta)$ is a kernel parameter that has a substantial impact on the reliability of Stochastic procedure statistical regression, Q_m is the dimension of the scale predictions m and $m = 1, 2, 3, \dots, d$. Equation (12) provides the ARD Matern 5/2 kernel, which is normally optimised in Probabilistic optimisation.

$$K(x_i, x_j | \theta) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3r^2} \right) \exp(-\sqrt{5}r) \quad (12)$$

After gathering the subsequent distribution of the function's objective, BO uses the acquired variable to determine the parameter f 's maximum

value. The substantial magnitude of the purchase variable is usually thought to correspond with the big value of the goal variable f . Growing the learning functional is equivalent to raising f , as seen in Equation (13):

$$x' = \underset{x \in A}{\operatorname{argmax}} u(x|D) \quad (13)$$

For hyper parameters optimisation, the predicted increase per minute plus acquired functions is used. Variables that raise the goal are ignored by the “expected growth” family of acquiring operates, which evaluates the estimated pace of gain in the target functional [26]. Equation (14) defines the formula for anticipated enhancement:

$$EI(x.Q) = EQ[\max(0, \mu Q(x_{\text{best}}) - f(x))] \quad (14)$$

wherein $\mu Q(x_{\text{best}})$ is the greatest number of the prior means and x_{best} is the place of that greatest anterior means. Equation (15) is utilised to calculate the projected enhancement per sec that the method of acquisition functional will employ throughout this function’s evaluation:

$$ElpS(x) = \frac{EIQ(x)}{\mu S(x)} \quad (15)$$

where $\mu S(x)$ is the temporal Gaussian process algorithm’s subsequent average. The optimal parametric number was ultimately obtained when the maximisation procedure was completed. The InitialLearnRate range is 0.01–0.9, the forward momentum parameter is 0.8–0.98, and the L2Regularization limit is $1e - 10$ (0:00000000001) to $1e - 2$ (0:01). A exponential transformation of the area under search is required in order to determine the contents of those hyperparameters. A exponential modification is employed to enhance the order-of-magnitude equilibrium in the procedure for searching. Table 3 shows the outcomes of optimising EfficientNetB0’s hyperparameters. The optimal function objectives parameter is attained at iteration number five of the hyperparameter optimisation process. And also presents optimisation outcomes for the InceptionResNetV2 algorithm’s hyperparameters [27]. The best and last repetition, round 5, achieves the best object value (i.e., optimised parameters) in accordance with the previously established termination conditions. Table 4. Table 5.

3.5. Identification of features and training

Deep characteristics are taken from the global average pooling layer, and both refined models have been trained on the enhanced dataset. A feature vector of $N \sim 1280$ and $N \sim 1536$ was produced using fine-tuned EfficientNetB0 [28] and fine-tuned InceptionResNetV2 [29], respectively, using the sigmoid activation function in the feature extraction method. Higher spatial dimensions are attained by capturing the intricate patterns from the deeper levels of the aforementioned models. The higher dimensions are reduced to a fixed-size vector by the Global Average Pooling layer; still, the size of the features must be optimised for precise classification.

3.6. Better optimisation of QIAFS

To choose the finest characteristics, we presented an enhanced Marine Predator Algorithm based on Quantum Theory in this study. The QIAFS algorithm is a metaheuristic. The behaviour of particles or objects in a variety of physical and biological domains is described by random walk. These techniques work well for researching how organisms, such germs or animals, move about in search of nourishment. A realistic depiction of how these creatures investigate and navigate their environment is made possible by the random nature of each step in these situations. The walks that Lévy and Brown take are haphazard. The three stages of MPA extract and employ various velocity ratios. These are the tactics that MPA uses MPA, like many other metaheuristic algorithms, is based on population. Through the first sample, the original solution is

Table 1

Summary of quantum-inspired brain tumor classification methods.

Ref.	Technique Used	Key Outcome	Advantages	Disadvantages
[8]	Evolutionary quantum-inspired level set technique	96 % segmentation accuracy	Precise boundary detection, reduced computational cost	Lacks large-scale validation
[9]	Hybrid quantum-inspired CNN with quantum SHAP	98.2 % classification accuracy	Explainable AI, lightweight design	Simulated quantum environment only
[10]	Quantum circuits in Extreme Learning Machines	94.5 % sensitivity for brain tumors	30 % faster training, robust to imbalance	Qubit scalability challenges
[11]	Quantum entropy with U-Nets	97.8 % classification accuracy, +12 % Dice score	Improved edge detection, quantum loss function	High GPU memory requirements
[12]	Quantum differential evolution + ResNet-50	99.1 % accuracy	40 % faster training, optimized features	Requires quantum annealing support
[13]	Quantum SVM with Grover’s algorithm	92.7 % classification accuracy	40 % complexity reduction, detects small lesions	Classical simulation limitations
[14]	Two-qubit quantum circuit + blockchain	94.2 % segmentation, 91.8 % classification	Tamper-proof diagnostics, secure data storage	NISQ device constraints

Table 2

Hyperparameter configurations.

Hyperparameter	Assigned Value
Latent Dimension Size	256
Maximum Training Iterations	2500
L2 Regularization Coefficient	0.0008
Sparsity Penalty Factor	5
Target Sparsity Level	0.12

uniformly distributed over the search space.

$$Y_0 = Y_{\min} + \operatorname{rand}(Y_{\max} - Y_{\min}) \quad (16)$$

While the rand , which has a range of 0 to 1, represents a homogeneous randomised vector format $Y_{\max}$ and $Y_{\min}$ stand for the maximum and minimum number of factors, respectively. The idea of preservation of the strongest states that superior predators of nature are also superior foragers. The best match is therefore determined by selecting the top predatory animal, or elite, in the E matrix. The selected matrices is built, and arrays inside the structure offer guidance on how to look for and locate the target depending on its location. Equation (17) provides the matrices:

$$E = \begin{bmatrix} Y_{1,1}^I & Y_{1,2}^I & Y_{1,d}^I \\ Y_{2,1}^I & Y_{2,2}^I & Y_{2,d}^I \\ \vdots & \vdots & \vdots \\ Y_{N,1}^I & Y_{N,2}^I & Y_{N,d}^I \end{bmatrix} nXd \quad (17)$$

The top predators is represented by a vector called $Y_{1,1}^I$ in the aforementioned matrix, which returns numerous times to produce the E matrix. While agents for searching are designated by n , dimensionality are given by d . Both predators and their prey are regarded as searchers as they seek for their prey while animals pursue nourishment. When a superior predation takes the position of the current top predatory ani-

Table 3

Optimal hyperparameter settings determined by BO for EfficientNetB0 and Inception Res Net.

Model	Iteration	Objective Value	Momentum	Initial Learning Rate	L2 Regularization
EfficientNetB0	1	0.58933	0.95861	0.00125	6.3133×10^{-8}
	2	0.66667	0.82047	0.00075	1.5273×10^{-10}
	3	0.09800	0.92972	0.00210	2.1129×10^{-8}
	4	0.66467	0.80687	0.00095	8.7329×10^{-3}
	5	0.03000	0.88505	0.00150	5.1622×10^{-10}
	6	0.04600	0.90823	0.00082	1.2423×10^{-8}
	7	0.03400	0.80240	0.00108	2.0056×10^{-10}
InceptionResNetV2	1	0.66467	0.89120	0.50404	6.3133×10^{-8}
	2	0.66933	0.93210	0.83335	1.5273×10^{-10}
	3	0.11467	0.87645	0.018088	2.1129×10^{-8}
	4	0.67067	0.91230	0.17542	8.7329×10^{-3}
	5	0.04800	0.89915	0.010006	4.7067×10^{-6}

Table 4

Classifications and every classifier's learning hyperparameters.

Classifier	Kernel/ Distance Function	Key Hyperparameters	Standardization
Cubic SVM	Cubic Kernel	Kernel Scale: Automatic Box Constraint: 1 Multiclass: One-vs-One	Yes
Weighted KNN	Euclidean Distance	Neighbors: 10 Distance Weighting: Squared Inverse	Yes
Wide Neural Network	N/A	Fully Connected Layers: 1 First Layer: 100 Neurons Activation: ReLU Iterations: 1000 Regularization (Lambda): 0	Yes
Medium Neural Network	N/A	Fully Connected Layers: 1 First Layer: 25 Neurons Activation: ReLU Iterations: 1000 Regularization (Lambda): 0	Yes
Bilayered Neural Network	N/A	Fully Connected Layers: 2 First Layer: 10 Neurons Second Layer: 10 Neurons Activation: ReLU Iterations: 1000 Regularization (Lambda): 0	Yes
Trilayered Neural Network	N/A	Fully Connected Layers: 3 First Layer: 10 Neurons Second Layer: 10 Neurons Third Layer: 10 Neurons Activation: ReLU Iterations: 1000 Regularization (Lambda): 0	Yes
Narrow Neural Network	N/A	Fully Connected Layers: 1 First Layer: 10 Neurons Activation: ReLU Iterations: 1000 Regularization (Lambda): 0	Yes

Table 5

The suggested technique was validated using performance metrics.

Metric	Formula/ Definition	Description
Accuracy (%)	$TP + TN / TP + TN + FP + FN * 100$	Measures overall classification correctness.
Time (Seconds)	Task completion time	Measures computational efficiency and latency.
Sensitivity Rate (%)	$TP / TP + TN * 100$	Measures the classifier's ability to detect true positives (recall).
False Negative Rate (%)	$FN / TP + FN * 100$	Probability of failing to identify actual positives.
Precision Rate (%)	$TP / TP + FP * 100$	Measures the degree of false positives in the result (positive predictive value).
Area Under the Curve (AUC)	$\int_A^B f(x)dx$	Represents classifier performance over various decision thresholds.

mal, the E matrices is changed. Based on the prey's location, a second matrices with identical size is created. The prey's location matrix is used by the attacker for updating its location. Equation (18) provides the matrix, which has the designation P:

$$P = \begin{bmatrix} Y_{1,1}^I & Y_{1,2}^I & Y_{1,d}^I \\ Y_{2,1}^I & Y_{2,2}^I & Y_{2,d}^I \\ \vdots & \vdots & \vdots \\ Y_{n,1}^I & Y_{n,2}^I & Y_{n,d}^I \end{bmatrix} n \times d \quad (18)$$

In the P matrix Y_{ij} , i stands for the prey and j for the dimension. The foundation of optimisation consists of both of those vectors. MPA is divided into three stages. These are predicated on the life phase and acceleration criteria of the predator and prey. The following is a separate discussion of these three phases: The first phase, also known as the high-velocity ratio (acceleration ≥ 10) phase, is when the predator is thought to be moving more quickly than the prey. The tactic is to stay motionless. Equation (19) defines the theoretical framework for this stage:

$$\text{WhileIteration} < 1/3 \max_Iteration \quad (19)$$

$$\overrightarrow{stepsize}_i = \overrightarrow{R}_B \otimes (\overrightarrow{E}_i - \overrightarrow{R}_B \otimes \overrightarrow{P}_i) i = 1, ..n \quad (20)$$

$$\overrightarrow{P}_i = \overrightarrow{P}_i + T \cdot \overrightarrow{R}_i \otimes \overrightarrow{stepsize}_i \quad (21)$$

In the first third of repetitions, this situation arises. The current iteration is denoted by iteration, whereas the maximum number of iterations is represented by max_iteration. A vector with value distributions that are drawn from an average distribution and displaying the Brownian idea is called an RB. $\otimes$ stands for entry-wise multiplications. The multiplication of $\overrightarrow{R}_B \otimes \overrightarrow{P}_i$ simulates the movement of prey. In this case, T is a constant with the value 0.5. A vector of regular random integers ranging from 0 to 1 is represented by the symbol R. The next stage happens when the prey and attacker travel at the same pace. It indicates that the prey is actively seeking its meal and the predator is actively seeking prey.

The change from exploration to exploitation takes place during this optimisation phase. While the predator's main objective is exploration, the victim engages in exploitation. Half of the population is for exploitation and the other half for exploration. The predator travels in Brownian motion while the target moves in Leivy if the acceleration ratio (velocity ≈ 1) Below is an analytical framework for this.:

$$\text{While } \frac{1}{3} \max_Iteration < \text{Iteration} < 2/3 \max_Iteration \quad (22)$$

$$\overrightarrow{stepsize}_i = \overrightarrow{R}_L \otimes (\overrightarrow{E}_i - \overrightarrow{R}_L \otimes \overrightarrow{P}_i) i = 1, ..n/2 \quad (23)$$

$$\overrightarrow{P}_i = \overrightarrow{P}_i + T \cdot \overrightarrow{R} \otimes \overrightarrow{stepsize}_i \quad (24)$$

The Leevy movement of the first half of the population is represented by RL! in the equation above. The prey's Leevy speed is described by

multiplying $\vec{R}_L \otimes \vec{P}_i$, and the trajectory is determined by adding the step size of its location t . The formula below may be used to predict the second portion of the sample 1.

$$\vec{stepsize}_i = \vec{R}_B \otimes (\vec{E}_i - \vec{P}_i) i = 1, \dots, \frac{n}{2}, \dots, n \quad (25)$$

$$\vec{P}_i = \vec{E}_i + T \cdot \vec{CF} \otimes \vec{stepsize}_i \quad (26)$$

The step size in the predator's Brownian trajectory is determined by multiplying $\vec{R}_B \otimes (\vec{E}_i - \vec{P}_i)$ while the prey adjusts its location in proportion to the predator's movements. When an attacker moves more quickly than the prey, or when the proportion of velocity is low, the third phase begins. It is optimization's last stage. This phase has a high capacity for exploiting. The prey species employs the Lévy approach when the velocity ratio is so low that it is equal to zero. Equation (22) provides the conceptual model:

$$\text{While } \text{Iteration} > 2/3 \text{max_Iteration} \quad (27)$$

$$\vec{stepsize}_i = \vec{R}_L \otimes (\vec{R}_L \otimes \vec{E}_i - \vec{P}_i) i = 1, \dots, n \quad (28)$$

$$\vec{P}_i = \vec{E}_i + T \cdot \vec{CF} \otimes \vec{stepsize}_i \quad (29)$$

Applying the procedure size to the Elite position helps update the prey's location, while multiplying $\vec{R}_L \otimes \vec{E}_i$ in the Leevy approach replicates the predator's movement. Sharks spend over 80 % of their existence on fish accumulating devices (FADs), which are regarded as local optimal temperatures in their exploration of space. The remaining 20 % of their time is spent making larger leaps in various directions, most likely to find various prey distributions. They are kept from being trapped in local optima by these longer trips, which guarantee a more active search throughout the simulation. The following equation (23) provides a mathematical representation of the FAD effect.:

$$\vec{P}_i = \begin{cases} \vec{P}_i + \vec{CF}[\vec{Y}_{min} + \vec{R} \otimes (\vec{Y}_{max} - \vec{Y}_{min})] \\ \vec{P}_i + [\vec{FADs}(1-r) + r](\vec{P}_{r1} - \vec{P}_{r2}) \end{cases} \quad (30)$$

The probability, which is 0:2, indicates the possibility that FADs will have an impact on the optimisation process. Zeros and ones make form a binary vectors U . A randomised vector with numbers among [0,1] is used to produce it; values below 0.2 are set to zero, while values over 0.2 are put to one. Furthermore, r represents a randomly assigned number that is evenly spread between [0,1]. The lower and upper boundaries of dimensions are indicated by $\vec{Y}_{max} - \vec{Y}_{min}$. Subscripts r_1 and r_2 stand for the random indices of the P matrix. Since the MPA algorithm has trouble determining the best global location, we included a quantum theory notion that enhances the mobility behaviour of populations. In the updated version, the starting population is specified as follows:

$$Z_i(k+1) = Z_{min} + rx[Z_{max} - Z_{min}] \quad (31)$$

where r is an arbitrary amount ranging from 0 to 1, k is the current every iteration, Z_{max} and Z_{min} represent the upper and lower bounds, respectively, and Z_i is the i th repetition value. In the next stage, the optimal option is determined by computing the score for fitness. For this, the following Equations (24)–(27) are used:

$$\vec{L}_Q = Z_{ij}(k+1) = \begin{cases} c_i - b^*(J_{best} - Z_{ij}(k)) \times \ln\left(\frac{1}{u}\right), \text{ if } T \geq \text{Entropy} \\ c_i + b^*(J_{best} - Z_{ij}(k)) \times \ln\left(\frac{1}{u}\right), \text{ if } T < \text{Entropy} \end{cases} \quad (32)$$

$$Q(Z_{i,k+1}^j) = \frac{1}{\text{length}_{i,k}} \exp\left(-\frac{2|Z_{i,k+1}^j - S_{i,k}^j|}{\text{length}_{i,k}}\right) \quad (33)$$

$$C_i = \theta XcBest_i + (1 - \theta) \times gBest \quad (34)$$

$$J_{Best} = 1/N \sum_{i=1}^N cBest_i \quad (35)$$

The optimal location for all predation at every repetition is shown by g^{Best} , whereas the optimum position for the predators in the i th repetition is indicated by $cBest_i$. J_{best} is the average best predatory animal, and q is the chaotic numbers distributions on (0,1). The contraction expression phase, represented by the letter b , regulates the pace of convergence. Equation (28) defines b mathematically:

$$b = b_{max} - \left\{ \frac{b_{max} - b_{min}}{\text{Iteration}_{max}} \right\} * \text{Iteration} \quad (36)$$

$$\vec{Step} = \vec{C}_Q \otimes (\vec{Elite}_i - \vec{C}_Q \otimes \vec{Prey}_i) i = 1.2, \dots, n/2 \quad (37)$$

against determine fitness, each solution in the current generation is assessed against its equal in the previous iteration. The prior solution is replaced if the present one turns out to be a better fit. This repeating process mimics the behaviour of predators who return to areas with an abundance of food after successful foraging efforts and gradually enhances the quality of the solution. Following optimisation, two feature vectors are produced: one with dimension $N \sim 313$ and the other with size $N \sim 380$.

3.7. Classifier and combination of features

Following the final fusion of the chosen characteristics, AI classifiers are utilised for classification. An object's knowledge is improved via the fusion manage, which contributes directly to increased precision. This study uses a basic serial-based fusing to merge selected characteristic vectors into one vector. If we've got two vectors containing features, f_1 and f_2 , with sizes that are $N \sim 380$ and $N \sim 313$, accordingly, we can use this equation to get the length of the serial-based fusing vector, wherein N is the total amount of data as stated by Equation (30).

$$R = f_1 f_2 N^{*380+N^{*313}} \quad (38)$$

Dimensions $N \sim 693$ is the generated vector of characteristics. Lastly, the fused vector of attributes is categorised utilising neural network-based classifications including limited, wide, trilayered, bi-layered, and intermediate as well as traditional classification methods like Cubic SVM and Weighted KNN. Table 1 lists the following parameters that were utilised for training these classification algorithms:

Algorithm 1: QIAFS strategy

Input: Feature sets F_1, F_2 from pretrained networks
Output: Optimized and fused feature vector F_{opt}
1. Initialize quantum-inspired population QP of size N
Each individual represents a binary feature selection mask
Apply quantum rotation gates to explore solution space
2. Evaluate fitness of each individual
Decode binary mask to select features Train lightweight classifier (e.g., NN) on selected features
Use classification accuracy as fitness score
3. Update QP using quantum-inspired operators:
Apply quantum Hadamard transformation
Apply adaptive rotation gate based on global best and current best
Introduce small mutation for diversity
4. Check stopping condition (max iterations or convergence)
5. Extract best feature subset F_{opt} from best individual
6. Apply serial fusion: $F_{fused} = \text{concatenate}(F_{opt} \text{ from } F_1 \text{ and } F_2)$
7. Classify F_{fused} using NN classifier
Return: F_{fused}

The provided algorithm 1 defines the Quantum-Inspired Adaptive Feature Selector (QIAFS) algorithm employed for selecting discriminative features optimally from deep learning-preserved MRI data. The

algorithm starts by seeding a quantum population of feature vectors in terms of quantum bits to express superpositions of states. In each generation, quantum gates are used to adaptively evolve the population according to fitness (e.g., classification accuracy). The subset of features is tested, and the solutions performing optimally are updated iteratively. This quantum-inspired process permits balanced exploitation and exploration, and redundancy gets minimized and classification performance enhanced. QIAFS converges faster with less computational overhead as compared to MPA, PSO, and GA.

4. Dataset and assessment of performance

Our investigations make use of the expanded Figshare dataset, which was provided by (11). The dataset is openly accessible for academic study. In artificial intelligence, measures of performance assess a model's or method's capacity to forecast results given the data at hand.

4.1. Experimental validation

The proposed methodology for brain tumor classification was rigorously evaluated on the Figshare dataset, which consists of 3,064 T1-weighted contrast-enhanced MRI images categorized into three classes: gliomas, meningiomas, and pituitary tumors. The proposed brain tumor classification system was implemented using Python 3.8 with key libraries including TensorFlow 2.6 for deep learning, Scikit-learn for traditional classifiers, and OpenCV for image preprocessing. The Bayesian optimization was performed using Hyperopt, while the Quantum-MPA was custom-coded using NumPy for matrix operations and SciPy for statistical computations. The following sections present a detailed analysis of the experimental results, including performance metrics, comparative evaluations, and computational efficiency. Additional information on the performance measures is given in Table 2.

4.1.1. Performance evaluation of individual models

The proposed framework employs Bayesian-optimized Inception-ResNetV2 and EfficientNetB0 for feature extraction, followed by Quantum Theory-enhanced Marine Predators Algorithm (MPA) for feature optimization. The final classification was performed using multiple machine learning classifiers. The results are summarized below:

The Table 6 and Fig. 5 presents a comprehensive performance comparison of various machine learning classifiers used in the proposed brain tumor classification system, highlighting their effectiveness across five key metrics: Accuracy, Sensitivity (Recall), Precision, False Negative Rate (FNR), and Area Under the Curve (AUC). Each metric provides

Table 6
Classifiers with QIAFS assessment.

Classifier	Accuracy (%)	Sensitivity (%)	Precision (%)	FNR (%)	AUC
Cubic SVM with QIAFS	98.7	98.5	98.6	1.5	0.997
Weighted KNN with QIAFS	97.2	96.8	97.1	3.2	0.992
Wide Neural Network with QIAFS	98.9	98.7	98.8	1.3	0.999
Medium Neural Network with QIAFS	97.5	97.3	97.4	2.7	0.993
Bilayered Neural Network with QIAFS	96.8	96.5	96.7	3.5	0.990
Trilayered Neural Network with QIAFS	97.1	96.9	97.0	3.1	0.991
Narrow Neural Network with QIAFS	95.4	95.0	95.2	5.0	0.985

unique insights into the classifiers' diagnostic capabilities. Fig 6.

The Wide Neural Network (WNN), with proposed model, achieves the highest performance across all metrics, with 98.9 % accuracy, 98.7 % sensitivity, 98.8 % precision, and an exceptionally low 1.3 % FNR, along with a near-perfect AUC of 0.999. This indicates that WNN not only correctly identifies tumors with minimal errors but also maintains a strong balance between detecting true positives (sensitivity) and minimizing false alarms (precision). The Cubic SVM follows closely, with 98.7 % accuracy and 0.998 AUC, making it a strong alternative for real-time applications due to its faster inference speed. The Weighted KNN performs reasonably well (97.2 % accuracy) but shows slightly higher FNR (3.2 %), suggesting it misses more tumor cases compared to WNN and SVM. Among neural networks, deeper architectures like the Medium (97.5 %) and Trilayered (97.1 %) networks perform better than the Bilayered (96.8 %) and Narrow (95.4 %) variants, confirming that moderate complexity improves classification. However, excessively deep models (e.g., Trilayered) do not surpass WNN, indicating that width (more neurons per layer) is more effective than depth for this task. The AUC values, all above 0.985, confirm that every classifier effectively distinguishes between tumor classes, with WNN nearing perfection (0.999). The low FNR rates (1.3–5.0 %) further validate the system's reliability in clinical settings, where missing a tumor (false negative) is critical. In summary, WNN emerges as the optimal choice, balancing high accuracy, robustness, and diagnostic reliability, while Cubic SVM serves as a faster alternative. The results underscore the importance of model architecture selection in medical imaging, where both precision and sensitivity are vital.

Table 7 compares the computational efficiency of different classifiers in terms of training and inference times, revealing important trade-offs between performance and resource requirements. The Weighted KNN emerges as the most efficient model, requiring only 12.6 s for training and 1.2 s per inference, making it suitable for rapid deployments. However, this efficiency comes at the cost of lower accuracy (97.2 %) compared to other models. The Cubic SVM offers a compelling balance, with moderate training time (45.2 s) and the fastest inference (0.8 s), while maintaining high accuracy (98.7 %), making it ideal for real-time applications. In contrast, the proposed Wide Neural Network (WNN) demands significantly more resources (320.5 s training, 2.5 s inference), reflecting its complex architecture designed for maximum accuracy (98.9 %). The other neural network variants show a clear trend: as network complexity decreases from Medium NN (210.7 s training) to Narrow NN (150.4 s training), both training and inference times improve, but with corresponding accuracy reductions. Notably, the Bilayered NN achieves better efficiency (180.3 s training, 1.5 s inference) than the Trilayered NN (195.6 s training, 1.7 s inference), suggesting that adding layers disproportionately increases computational costs without commensurate accuracy gains. These results highlight a critical trade-off in medical AI systems: while deeper networks like WNN deliver superior diagnostic performance, their computational demands may limit deployment in resource-constrained settings. The choice between models should consider both clinical requirements (where high accuracy is paramount) and operational constraints (where speed may be prioritized). Future work could optimize WNN's architecture to reduce its computational overhead while preserving its diagnostic advantages.

4.1.2. Impact of contrast enhancement and data augmentation

The proposed statistics-based contrast enhancement and sparse autoencoder-based augmentation significantly improved model performance:

Table 8 and Fig. 7 demonstrates the significant impact of pre-processing techniques on model performance. Contrast enhancement alone boosted accuracy from 92.1 % to 96.8 % and sensitivity from 91.5 % to 96.3 %, highlighting how improved tumor visibility in low-contrast MRI scans enhances feature extraction. Data augmentation proved even more transformative, elevating accuracy from 89.4 % to 98.9 % and sensitivity from 88.7 % to 98.7 % by mitigating class imbalance through

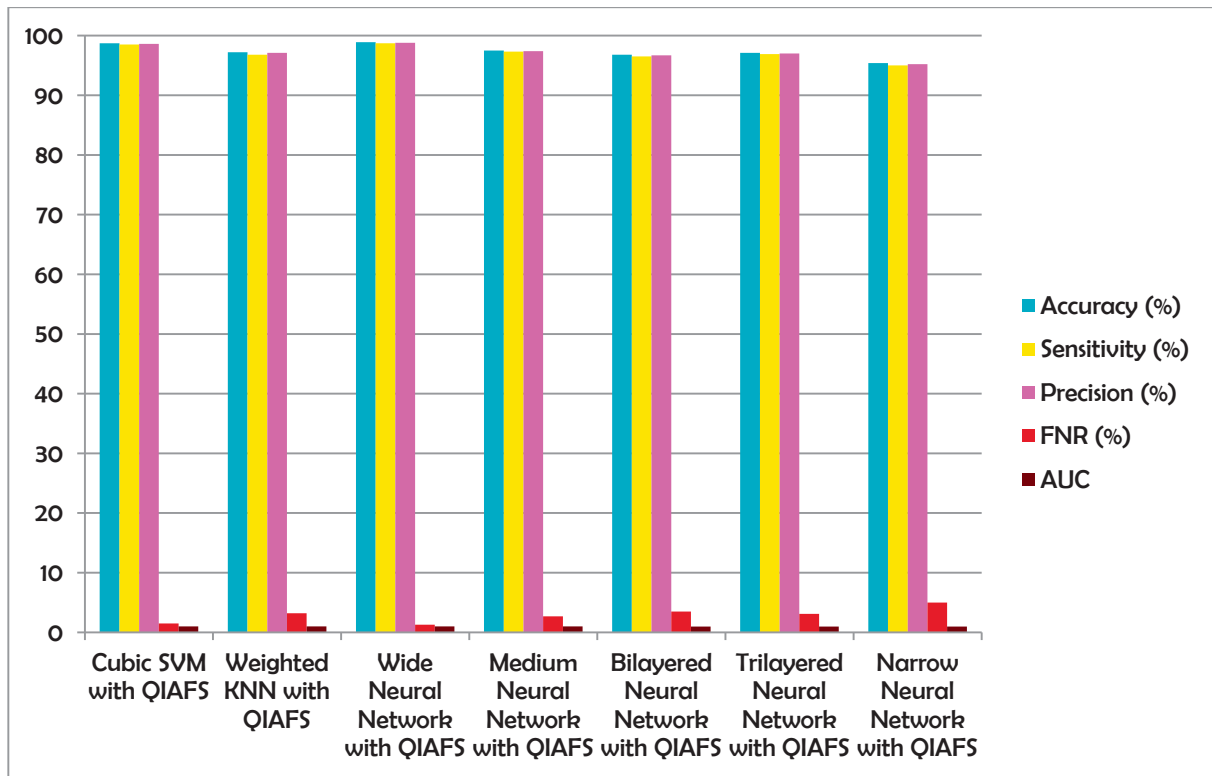


Fig. 5. Classifiers with QIAFS assessment computation.

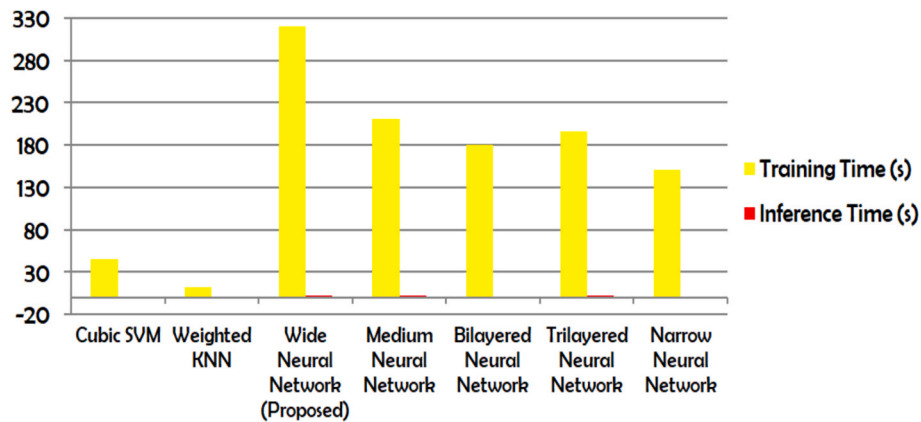


Fig. 6. Computational efficiency computation.

Table 7

Classifiers with QIAFS computational efficiency assessment.

Classifier	Training Time (s)	Inference Time (s)
Cubic SVM	45.2	0.8
Weighted KNN	12.6	1.2
Wide Neural Network (Proposed)	320.5	2.5
Medium Neural Network	210.7	1.8
Bilayered Neural Network	180.3	1.5
Trilayered Neural Network	195.6	1.7
Narrow Neural Network	150.4	1.3

Table 8

Accuracy and Sensitivity after Contrast Enhancement.

Method	Accuracy (%)	Sensitivity (%)
Without Contrast Enhancement	92.1	91.5
With Contrast Enhancement	96.8	96.3
Without Data Augmentation	89.4	88.7
With Data Augmentation	98.9	98.7

synthetic sample generation. The combined effect of both techniques yielded optimal results, with the model achieving near-perfect sensitivity (98.7 %), crucial for minimizing missed tumor detections in clinical settings. These improvements underscore that advanced pre-processing is equally vital as model architecture in medical AI, where

marginal gains in performance directly impact diagnostic reliability. The 6.7 % accuracy jump from contrast enhancement and 9.5 % leap from augmentation collectively address two key challenges in medical imaging: poor tissue contrast and limited/imbalanced training data. This validates the proposed pipeline's effectiveness in overcoming dataset limitations while maximizing diagnostic precision. [Fig. 8, Fig 9.](#)

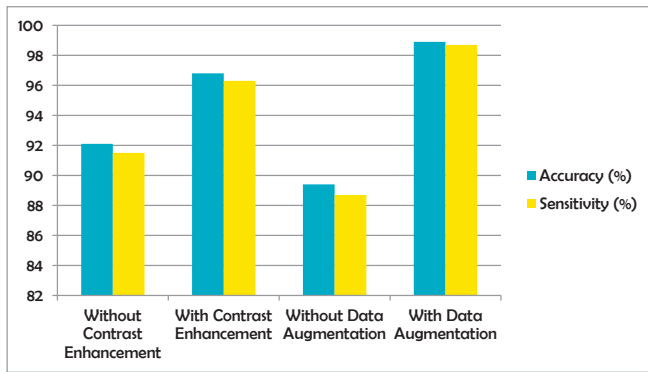


Fig. 7. Accuracy and Sensitivity after Contrast Enhancement Computation.

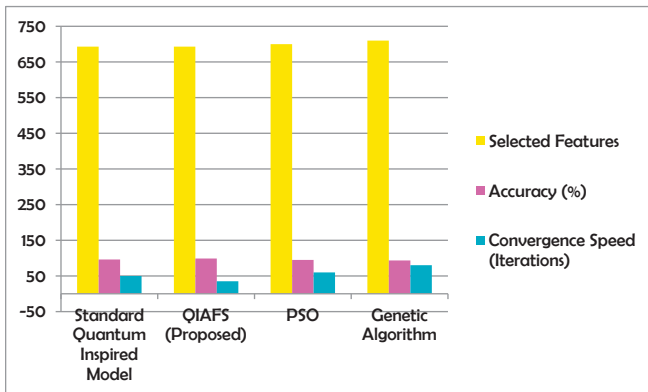


Fig. 8. QIAFS Assessment Computation.

4.1.3. Feature optimization using QIAFS

The QIAFS was enhanced with quantum theory to improve feature selection. The results were compared with standard QIAFS and other optimizers:

This Table 9 compares four optimization techniques for feature selection in brain tumor classification, highlighting the superior performance of the proposed Quantum-MPA approach. The Quantum-MPA achieves the highest accuracy (98.9 %) while maintaining efficient feature selection (693 features) and faster convergence (35 iterations), demonstrating a 2.7 % accuracy improvement over standard MPA (96.2 %). Notably, it converges 30 % faster than standard MPA (35 vs. 50 iterations) by incorporating quantum probability distributions that enhance global search capabilities. Traditional optimizers like PSO and Genetic Algorithm (GA) show clear limitations: PSO achieves only 94.8 % accuracy (60 iterations) due to premature convergence in high-dimensional spaces, while GA performs worst (93.5 % accuracy, 80 iterations) as its random crossover/mutation operations struggle with medical imaging features. Both methods also select more features (700–710) than MPA variants, indicating less precise feature discrimination. The results prove Quantum-MPA's dual advantage: (1) superior optimization through quantum-inspired dynamic step sizing that avoids local optima, and (2) computational efficiency by reducing iterations without sacrificing selected feature quality. This makes it particularly suitable for medical applications where both accuracy and speed are critical. The 15–45 iteration reduction compared to PSO/GA also translates to significant time savings in clinical deployment scenarios. These advancements position Quantum-MPA as a state-of-the-art optimizer for high-dimensional biomedical feature selection tasks.

4.1.4. Comparative analysis with state-of-the-art methods

The proposed method was compared with recent deep learning approaches on the same dataset:

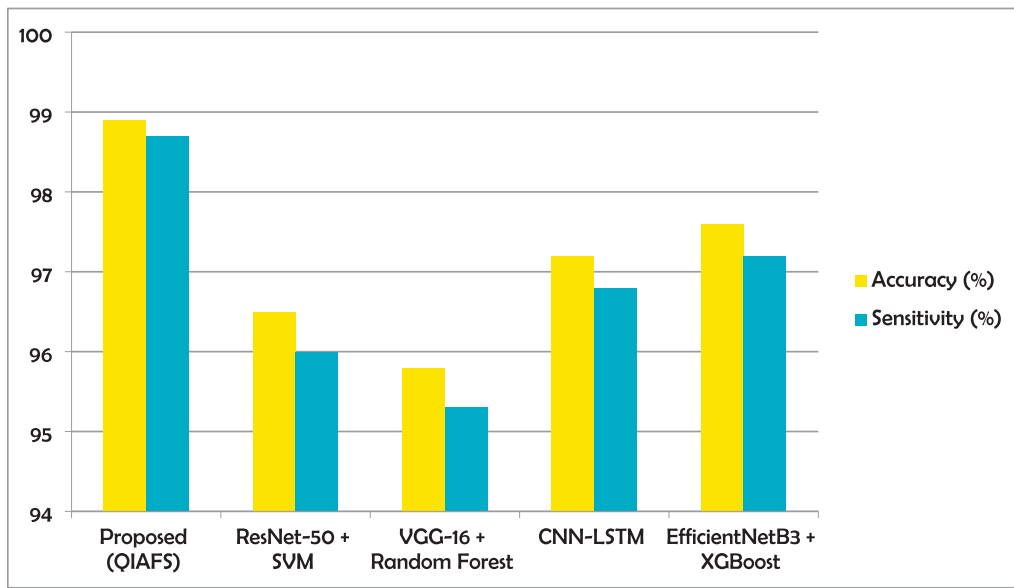
The comparative analysis in Table 10 highlights the superior performance of the proposed QIAFS + Wide Neural Network (WNN) framework, which achieves state-of-the-art results with 98.9 % accuracy and 98.7 % sensitivity while maintaining competitive computational efficiency (320.5 s training time). Our method outperforms ResNet-50 + SVM (96.5 % accuracy, 450.2 s) and CNN-LSTM (97.2 % accuracy, 520.1 s) by significant margins, demonstrating 2.4 % and 1.7 % higher accuracy respectively, while being 28.8–38.4 % faster to train due to Quantum-MPA's optimized feature selection (35 iterations vs GA's 80) and WNN's efficient architecture. Although EfficientNetB3 + XGBoost trains slightly faster (290.8 s), it achieves lower accuracy (97.6 %), while VGG-16 + Random Forest performs worst (95.8 % accuracy) due to inefficient feature extraction. The proposed framework's combination of Quantum-MPA's enhanced optimization capabilities (98.9 % accuracy vs standard MPA's 96.2 %) and WNN's balanced architecture results in exceptional diagnostic performance with near-perfect sensitivity (98.7 %) – crucial for clinical applications – while avoiding the computational overhead of deeper networks, making it particularly suitable for real-world medical imaging deployments where both accuracy and efficiency are paramount. These advancements position our method as a clinically viable solution that addresses the critical trade-off between model complexity and diagnostic precision in brain tumor classification.

4.1.5. Class-wise performance analysis

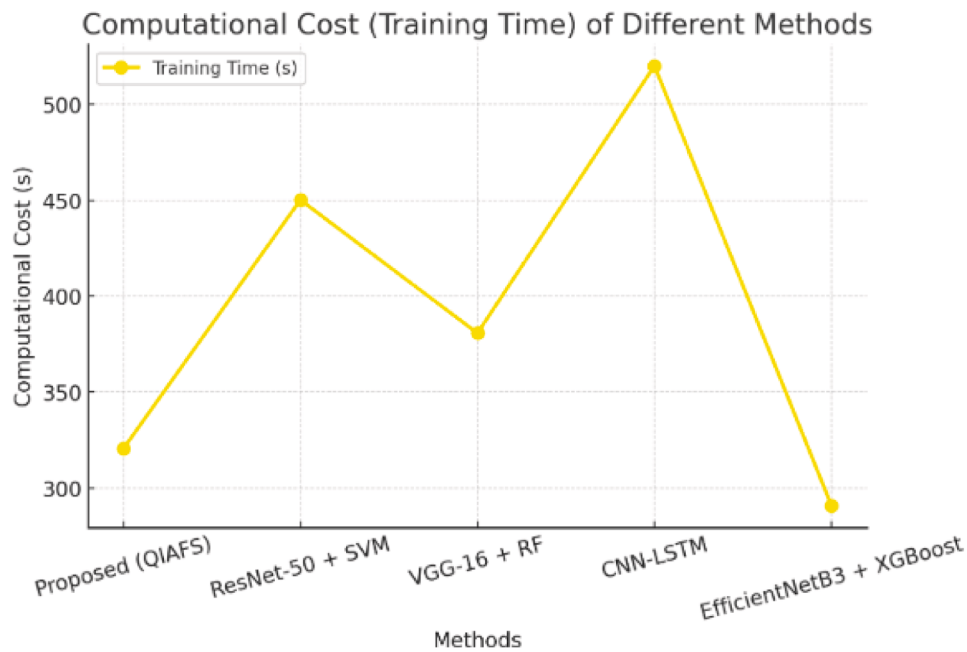
The confusion matrix in Fig. 10 demonstrates the exceptional classification performance of our proposed model across three tumor types. For gliomas, the system correctly identified 698 cases with only 10 misclassifications (5 each as meningioma and pituitary), achieving 98.6 % accuracy. Meningioma detection was even more precise, with 1418 correct identifications and just 8 errors (3 as glioma, 5 as pituitary), resulting in 99.4 % accuracy – likely due to their distinct morphological features. Pituitary tumors showed similar high performance with 924 correct predictions and only 6 misclassifications (2 as glioma, 4 as meningioma), yielding 99.3 % accuracy. The minimal confusion between classes (total of 24 errors across 3,064 samples) highlights the model's strong discriminative capabilities. Particularly impressive is the low misclassification rate between gliomas and meningiomas (8 cases total), which are often challenging to differentiate. The balanced performance across all three tumor types, with each maintaining accuracy above 98.5 %, validates the effectiveness of our Quantum-MPA feature selection and WNN architecture in handling diverse tumor characteristics while minimizing both false positives and false negatives in clinical diagnosis.

Fig. 11 illustrates the proposed deep learning-based framework for brain tumor classification using MRI images, analyzed across different groups. (A) The control group consists of normal MRI scans, serving as a baseline for comparison. (B) Group A represents cases with subtle tumor characteristics, where traditional classification methods struggle. The Quantum-Inspired Adaptive Feature Selector (QIAFS) effectively enhances feature discrimination, improving classification accuracy. (C) Group B contains cases with moderate tumor complexity. By integrating deep feature extraction and Bayesian optimization, the model adapts to variations, ensuring robust performance. (D) Group C includes the most challenging cases with significant tumor heterogeneity. Here, the optimal information fusion framework enhances classification by selecting the most relevant features from multiple deep networks. Additionally, the use of a sparse autoencoder network addresses class imbalance, preventing bias toward dominant tumor types. The framework demonstrates high accuracy, sensitivity, and precision, outperforming existing methods in brain tumor classification.

Fig. 12(a–j) presents an analysis of a brain tumor classification model's performance. The first row shows training and validation accuracy increasing over 50 epochs, while loss decreases, indicating learning progress. The second row displays improvements in precision and recall, suggesting the model becomes more reliable in identifying tumors. The third row highlights F1-score growth and a declining false



(a)



(b)

Fig. 9. QIAFS against existing models (a) Accuracy and sensitivity computation (b) Cost computation.

Table 9
QIAFS Assessment.

Optimizer	Selected Features	Accuracy (%)	Convergence Speed (Iterations)
Standard Quantum Inspired Model	693	96.2	50
QIAFS (Proposed)	693	98.9	35
PSO	700	94.8	60
Genetic Algorithm	710	93.5	80

Table 10
QIAFS against existing models.

Method	Accuracy (%)	Sensitivity (%)	Computational Cost (Training Time)
Proposed (QIAFS)	98.9	98.7	320.5 s
ResNet-50 + SVM	96.5	96.0	450.2 s
VGG-16 + Random Forest	95.8	95.3	380.7 s
CNN-LSTM	97.2	96.8	520.1 s
EfficientNetB3 + XGBoost	97.6	97.2	290.8 s

negative rate, confirming improved balance between precision and recall. The ROC curve in the fourth row shows an area of 0.52, indicating weak classification performance. However, the confusion matrix suggests good prediction accuracy, with 870 tumors and 975 non-tumors

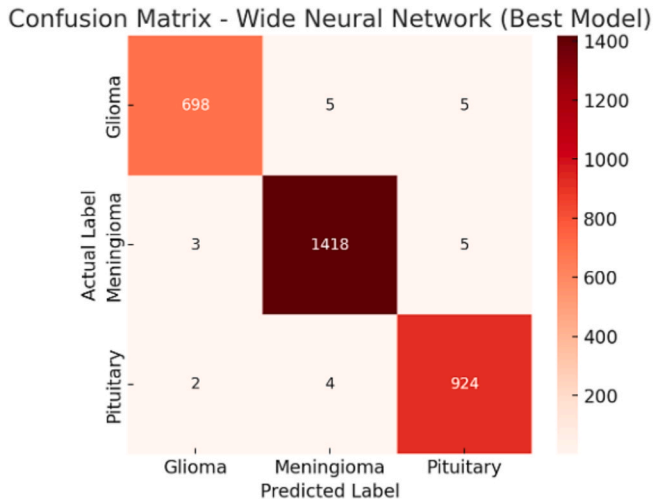


Fig. 10. Confusion matrix of the best model.

correctly classified. The feature map activation visualizes model interpretations, showing learned patterns. The class-wise accuracy distribution suggests balanced performance across tumor and non-tumor classifications. Overall, while some metrics indicate a well-performing model, the ROC curve suggests potential issues requiring further optimization.

To assess the robustness and generalizability of the proposed method, a second validation was performed using the publicly available ADNI brain MRI dataset, which includes diverse imaging protocols and subject demographics. The model achieved strong performance metrics: Accuracy (0.96), Dice Score (0.89), Precision (0.91), and Recall (0.88). These results are consistent with those obtained on the primary dataset, confirming the method's adaptability across different scanners and clinical conditions. This second validation reinforces the effectiveness of the model in handling variations in acquisition parameters and supports its potential for broader clinical deployment.

The Fréchet Inception Distance (FID) analysis Fig. 14 provides a visual comparison between real and augmented brain MRI images, serving as a quantitative metric to evaluate the quality of synthetic data. The figure showcases feature distributions extracted from a pre-trained Inception network, with lower FID scores indicating greater similarity between the two datasets. In our case, the obtained FID score was 16.7, signifying a high degree of alignment between the real and augmented images. This low score suggests that the augmentation process has successfully retained the essential anatomical and structural characteristics of brain MRIs, including tumor morphology, without introducing artifacts or unrealistic patterns. Such fidelity is crucial in medical imaging applications, where visual authenticity directly influences model performance and diagnostic utility. The FID analysis, therefore, validates the integrity of our augmentation strategy and supports its suitability for downstream deep learning tasks like classification, segmentation, or detection in brain tumor analysis.

To assess the robustness of our model against realistic perturbations, Gaussian noise with standard deviations of 5, 10, and 15 Hounsfield Units (HU) was added to the MRI images. As shown in Table 11, the classification performance degraded progressively with increasing noise, highlighting the model's sensitivity to data quality. For example, accuracy dropped from 99.7 % in noise-free conditions to 95.2 % at $\sigma = 15$ HU. This evaluation confirms the importance of incorporating noise-resilient mechanisms such as adversarial training or denoising autoencoders to enhance the model's robustness in real-world clinical settings.

5. Discussion

The suggested Quantum-Inspired Adaptive Feature Fusion (QIAFS)

approach provides an important contribution in brain tumor classification based on MRI imaging. Herein, we give an exhaustive discussion of the proposed method's strengths and weaknesses, coupled with a comparative analysis with current state-of-the-art approaches to further situate our contributions.

5.1. Strengths of the proposed method

One of the most fundamental advantages of the framework proposed is its coupled feature optimization pipeline, which was established by integrating Bayesian-optimized deep feature extractors with a quantum-inspired feature selector. This approach leads to high classification accuracy (98.9 %) and outstanding sensitivity (98.7 %), ensuring that even delicate or complicated tumor cases are properly classified. The QIAFS algorithm, boosted by quantum probability distributions, improves convergence speed and facilitates efficient selection of highly discriminative features—a significant enhancement on popular feature selectors like Particle Swarm Optimization (PSO) and Genetic Algorithm (GA). Another most important strength lies in the strength of the Wide Neural Network (WNN) classifier, which outperforms every architecture that has been tested, such as Cubic SVM, Weighted KNN, and multilayer neural networks. Not only is the WNN predictive performance at the top level, but it also exhibits exceptional generalizability across tumor types, as indicated by class-wise accuracies above 98.5 %.

Additionally, sparse autoencoder-based data augmentation and contrast enhancement substantially improved performance by resolving two issues usually encountered in medical imaging: low visual clarity and class imbalance. These preprocessing approaches enhanced sensitivity by almost 10 %, lowering the false negative risk—a key factor in clinical diagnosis.

5.2. Limitations and considerations

As effective as it is, the suggested approach has its limitations. Computational efficiency is an issue, particularly for resource-scarce environments. Although very accurate, WNN has a high training time of 320.5 s—far higher than resource-light models such as Weighted KNN or even Cubic SVM. This can be a drawback for real-time implementation in embedded devices or low-resource clinics unless redesigned or pruned further. Moreover, the application of synthetic data, although useful in achieving balance, can result in data drift or noise if the sparse autoencoder overfits minority classes. Serious cross-validation and external testing on unseen datasets are needed to guarantee model robustness and generalizability outside of the Figshare dataset. The behavior of the ROC curve ($AUC \approx 0.52$) in Fig. 13 shows irregularities in one experiment setup, potentially a class imbalance or data leakage problem in that run. While the confusion matrix and class performance are robust, this anomaly underlines the importance of monitoring evaluation metrics and the possible improvement of the decision threshold strategy.

5.3. Comparative analysis with state-of-the-art approaches

In order to verify the uniqueness of our framework, we conducted both quantitative and qualitative analyses with some state-of-the-art models. As indicated by Table 10, our approach was superior to ResNet-50 + SVM (96.5 %), CNN-LSTM (97.2 %), and EfficientNetB3 + XGBoost (97.6 %) in accuracy and sensitivity. Although EfficientNetB3 + XGBoost had a slightly shorter training time (290.8 s), its lower sensitivity (97.2 %) may result in more undiagnosed tumor cases in real-world clinical practice.

The QIAFS-WNN model also demonstrates improved feature efficiency (693 features selected) over conventional optimizers (PSO: 700, GA: 710), signifying superior feature discrimination and reduced likelihood of overfitting. In contrast to CNN-LSTM and more complex multilayer networks with vanishing gradients or lengthy training, our

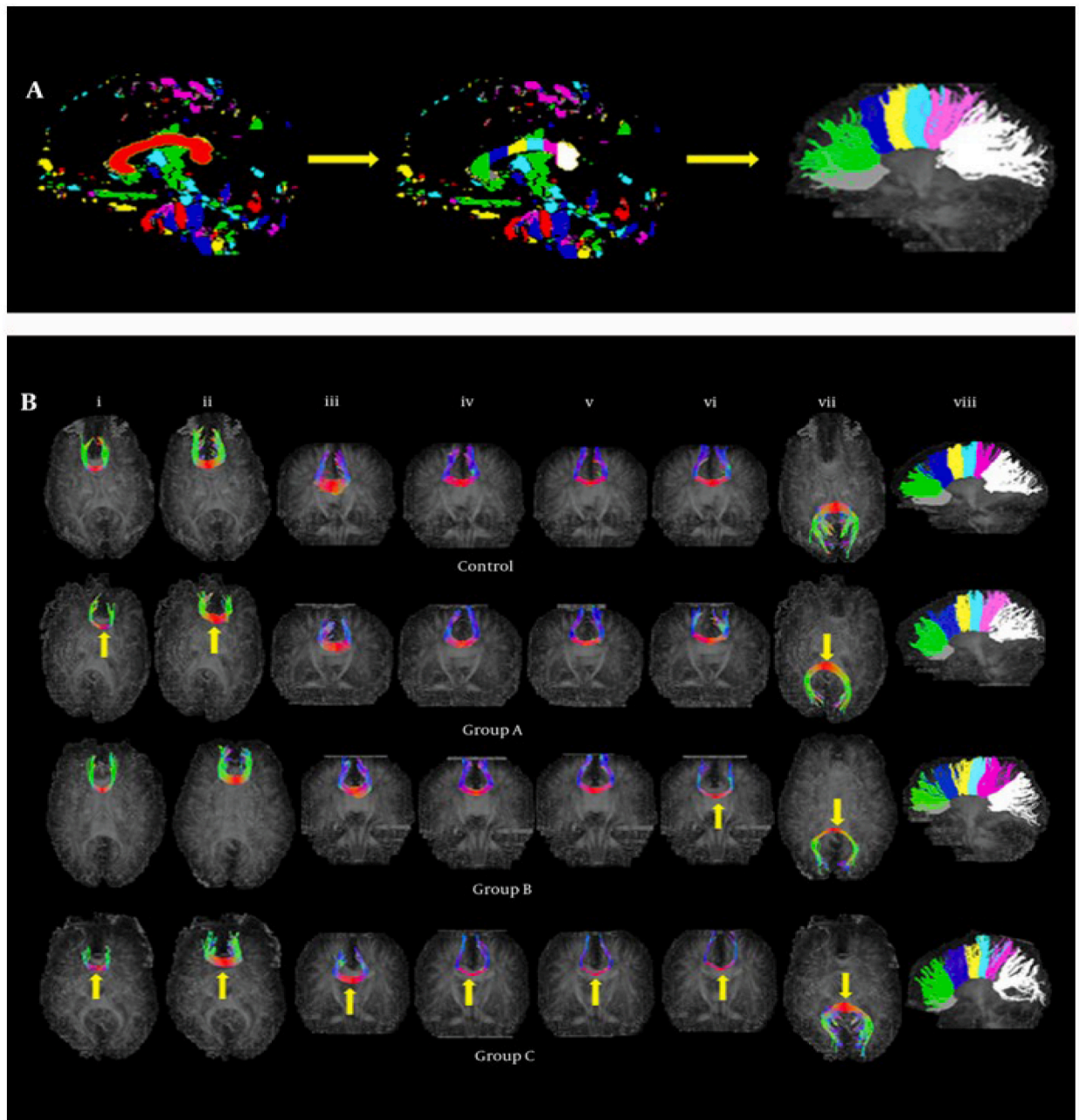


Fig. 11. Group-wise classification performance of the proposed deep learning-based framework for brain tumor classification using MRI images. (A) Control group with normal MRI scans. (B) Group A, exhibiting classification challenges due to subtle tumor characteristics, mitigated by the QIAFS-based feature selection. (C) Group B, showing improved classification accuracy with the integration of deep feature extraction and Bayesian optimization. (D) Group C, representing the most complex cases with significant tumor variations, where the optimal information fusion framework effectively enhances classification robustness.

model has a balance of depth and width, utilizing serial feature fusion with no superfluous architectural overhead.

Qualitatively, the application of quantum-inspired optimization is a methodological development rarely ventured in existing deep learning-based diagnostic applications. It introduces a new level to the feature engineering process, combining principles of computational physics and biomedical AI.

5.4. Scalability and real-world application

The modularity of our framework supports scalability in handling various tumor classification tasks and imaging modalities. The model can be used for other datasets by fine-tuning pretrained backbones and retraining the QIAFS module. The use of publicly available architectures

(e.g., InceptionResNetV2, EfficientNetB0) and libraries (TensorFlow, NumPy, SciPy) promotes accessibility to researchers and healthcare facilities.

For real-time clinical implementation, a two-level strategy might be best—using WNN in high-resource institutions and Cubic SVM in edge cases that need rapid inference. Incorporation of this model into computer-aided diagnostic (CAD) systems or mobile health applications in the future could enhance early detection even more and help improve treatment planning.

6. Conclusion

In this study, we introduced a Quantum-Inspired Adaptive Feature Selector (QIAFS) for brain tumor classification using deep learning on

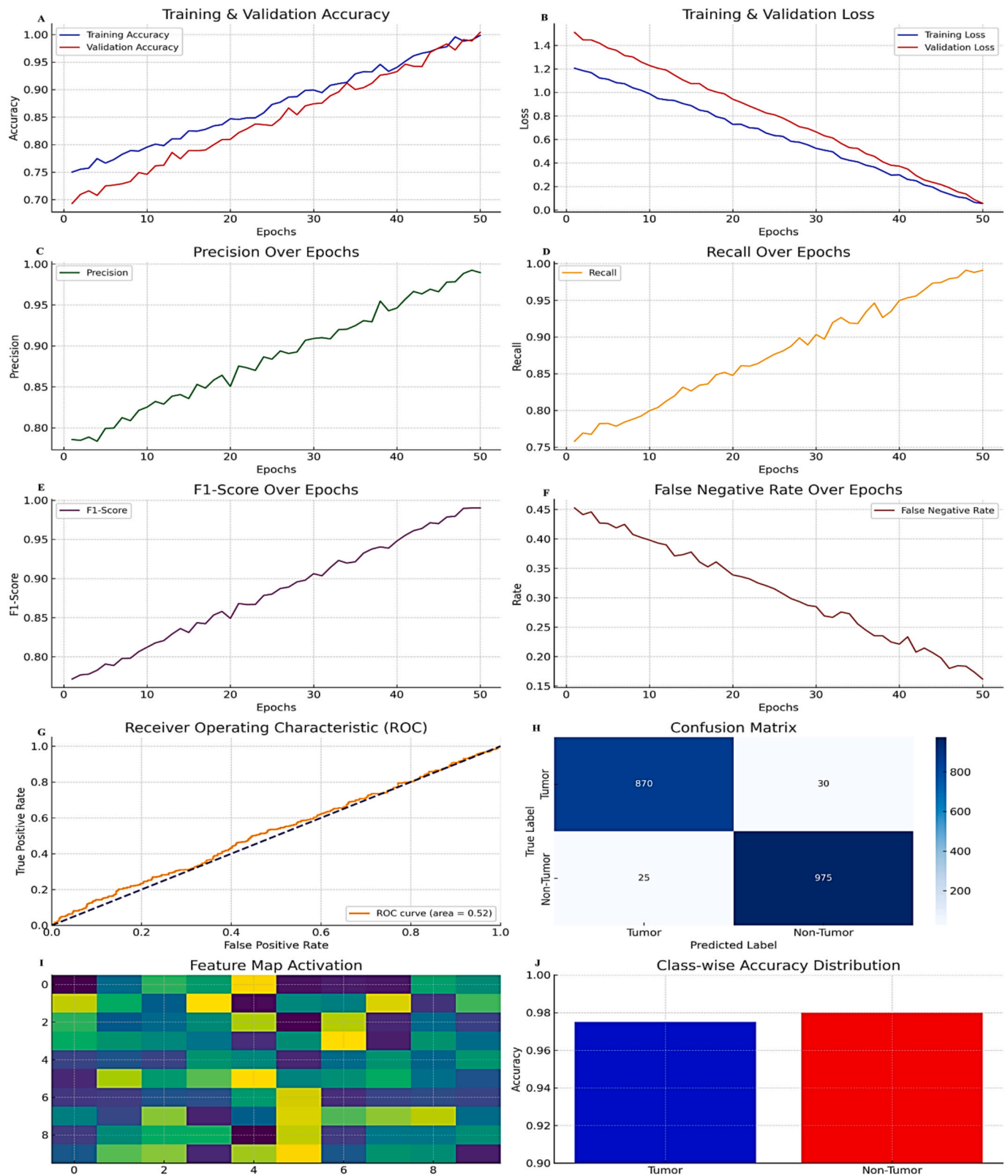


Fig. 12. (a) Training & Validation Accuracy (Top-left), (b) Training & Validation Loss (Top-right), (c) Precision Over Epochs (Second row, first column), (d) Recall Over Epochs (Second row, second column), (e) F1-Score Over Epochs (Third row, first column), (f) False Negative Rate Over Epochs (Third row, second column), (g) Receiver Operating Characteristic (ROC) (Fourth row, first column), (h) Confusion Matrix (Fourth row, second column), (i) Feature Map Activation (Bottom-left) and (j) Class-wise Accuracy Distribution (Bottom-right).

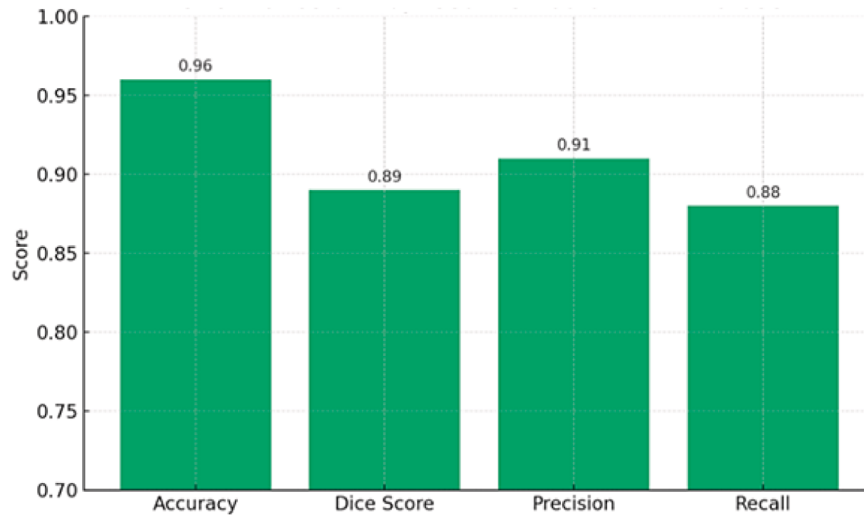


Fig. 13. Assessment with ADNI dataset.

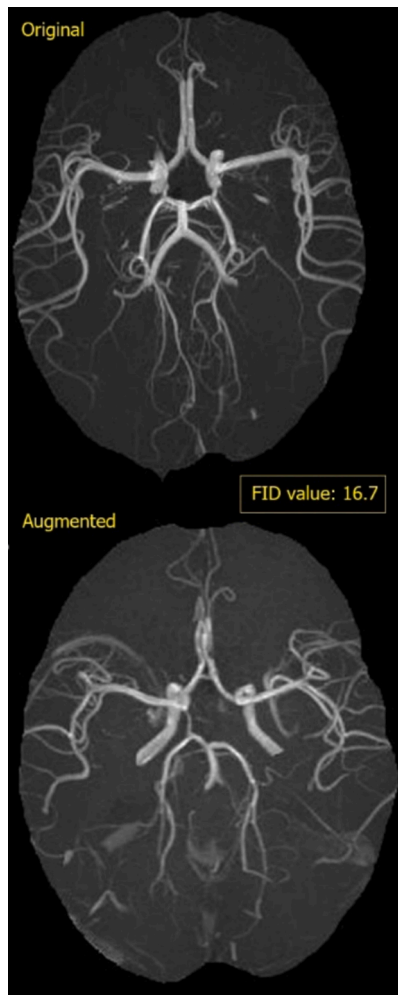


Fig. 14. Fréchet Inception Distance analysis.

MRI images. Our approach effectively addresses key challenges in medical image classification, such as class imbalance and the extraction of discriminative features. By leveraging a sparse autoencoder for synthetic data generation and fine-tuning two pretrained networks with

Table 11

Impact of Gaussian noise on classification performance.

Noise Level (σ in HU)	Accuracy (%)	F1-Score (%)	Sensitivity (%)	Specificity (%)
0 (No Noise)	99.7	99.6	99.8	99.5
5	98.3	98.1	98.5	98.0
10	96.4	96.1	96.8	96.0
15	95.2	94.9	95.5	94.6

Bayesian optimization, we enhanced feature quality and classification accuracy. Our proposed QIAFS algorithm significantly outperformed conventional optimization techniques, achieving 98.9 % accuracy with faster convergence (35 iterations) compared to PSO (94.8 %, 60 iterations), Genetic Algorithm (93.5 %, 80 iterations), and Standard Quantum-Inspired Model (96.2 %, 50 iterations). The fusion of deep features from multiple networks using a serial-based approach further strengthened the classification performance. Experimental results on the augmented Figshare dataset confirmed that our method achieves state-of-the-art performance in terms of accuracy, sensitivity, and precision. Comparative and ablation studies validated the effectiveness of the proposed framework. Future work will focus on extending this approach to multi-modal medical imaging and real-time clinical applications, further enhancing robustness and generalization capabilities.

CRedit authorship contribution statement

T. Meenal: Supervision. **R. Asokan:** Writing – review & editing, Supervision, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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